

1. Consider the following cost matrix for a transportation problem in which the objective is to minimize the total transportation cost.

Source	Destination			Supply
	1	2	3	
1	8	5	4	50
2	6	8	9	20
Demand	10	20	40	

- (a) (5 points) Write down the linear programming formulation for this problem.

→ Let x_{ij} be # of shipped units from Source i to Dest. j

Objective

$$\min Z = 8x_{11} + 5x_{12} + 4x_{13} + 6x_{21} + 8x_{22} + 9x_{23}$$

s.t.

Supply (constraints)

$$\rightarrow x_{11} + x_{12} + x_{13} = 50 \quad (\text{Source 1})$$

$$\rightarrow x_{21} + x_{22} + x_{23} = 20 \quad (\text{Source 2})$$

Demand (constraints)

$$x_{11} + x_{21} = 10 \quad (\text{Destination 1})$$

$$x_{12} + x_{22} = 20 \quad (\text{Dest. 2})$$

$$x_{13} + x_{23} = 40 \quad (\text{Dest. 3})$$

(b) (5 points) Set up the transportation tableau and use the northwest corner rule to find an initial basic feasible solution.

Source	Destination			Supply
	1	2	3	
1	8	5	4	50
2	6	8	9	20
Demand	10	20	40	

a.) (S_1, D_1) Supply = 50 Demand = 10

$$\rightarrow \min(10, 50) = 10$$

$$\text{new} \Rightarrow S_1 = 40, D_1 = 0$$

b.) (S_1, D_2) Supply = 40 Demand = 20

$$\rightarrow \text{alloc} \quad \min(20, 40) = 20$$

$$\text{new} \Rightarrow S_1 = 20, D_2 = 0$$

c.) (S_1, D_3) Supply = 20 Demand = 20

$$\text{new} \Rightarrow D_3 = 20$$

Initial Basic Feasible Solution:

Source	Dest.			Supply
	1	2	3	
1	10	20	20	50
2	0	0	20	20
Demand	10	20	40	

of basic vars = $2 + (3 - 1) = 4$ non-zero entries

(c) (5 points) Write down the dual problem of the linear program in part (a).

Defining Dual vars: u_1, u_2 (Source Constraints)
 v_1, v_2, v_3 (Dest. Constraints)

Dual Simplex Form

$$\Rightarrow \max Z = 50u_1 + 20u_2 + 10v_1 + 20v_2 + 40v_3$$

s.t.

$$u_1 + v_2 \leq 8$$

$$u_1 + v_2 \leq 5$$

$$u_1 + v_3 \leq 4$$

$$u_2 + v_1 \leq 6$$

$$u_2 + v_2 \leq 8$$

$$u_2 + v_3 \leq 9$$

and

$$u_i + v_j \leq c_{ij} \text{ for all } (i,j)$$

(d) (5 points) Write down all of the complementary slackness conditions.

$$\rightarrow x_{ij} (c_{ij} - u_i - v_j) = 0$$

Init. Basic
Feasible Sol.º

Source	Dest.		
	1	2	3
1	10	20	20
2	0	0	20

For each basic var.º

$$X_{11} = 10 \Rightarrow C_{11} = u_1 + v_1$$

$$u_1 + v_1 = 8$$

$$X_{12} = 20 \Rightarrow C_{12} = u_1 + v_2$$

$$u_1 + v_2 = 5$$

$$X_{13} = 20 \Rightarrow C_{13} = u_1 + v_3$$

$$u_1 + v_3 = 4$$

Thus...

$$X_{23} = 20 \Rightarrow C_{23} = u_2 + v_3$$

$$u_2 + v_3 = 9$$

$$\Rightarrow v_1 = 8, v_2 = 5, v_3 = 4, u_2 = 5$$

For non-basic vars:

$$\text{need } \rightarrow C_{ij} > u_i + v_j$$

$$\bullet \quad x_{21} = 0 \rightarrow u_2 + v_1 < 6$$

$$\bullet \quad x_{22} = 0 \rightarrow u_2 + v_2 < 8$$

All Complementary Slackness Conditions

$$\bullet \quad x_{11} (8 - u_1 - v_1) = 0$$

$$\bullet \quad x_{12} (5 - u_1 - v_2) = 0$$

$$\bullet \quad x_{13} (4 - u_1 - v_3) = 0$$

$$\bullet \quad x_{23} (9 - u_2 - v_3) = 0$$

$$\bullet \quad x_{21} (6 - u_2 - v_1) = 0 \Rightarrow u_2 + v_1 < 6 \quad \text{since } x_{21} = 0$$

$$\bullet \quad x_{22} (8 - u_2 - v_2) = 0 \Rightarrow u_2 + v_2 < 8 \quad \text{since } x_{22} = 0$$

(e) (10 points) Solve the problem using the transportation algorithm.

Init. Basic
Feasible Sol.°

Source	Dest.			Supply
	1	2	3	
1	10	20	20	50
2	0	0	20	20
Demand	10	20	40	

⇒ Basic Vars.° $x_{11} = 10$

$$x_{12} = 20$$

$$x_{13} = 20$$

$$x_{23} = 20$$

→ 4 basic vars.

° Computing Dual Vars (u_i, v_j)

Using° $C_{ij} = u_i + v_j$ (basic vars. only)

Let $u_1 = 0$

then...

$$\rightarrow x_{11} \Rightarrow 8 = u_1 + v_1 \Rightarrow v_1 = 8$$

$$\rightarrow x_{12} \Rightarrow 5 = u_1 + v_2 \Rightarrow v_2 = 5$$

$$\rightarrow x_{13} \Rightarrow 4 = u_1 + v_3 \Rightarrow v_3 = 4$$

$$\rightarrow x_{23} \Rightarrow 9 = u_2 + v_3 = u_2 + 4 \Rightarrow u_2 = 5$$

Dual Values : $u_1 = 0, u_2 = 5$
 $v_1 = 8, v_2 = 5, v_3 = 4$

• Check for optimality

$$\rightarrow x_{21} : 6 - 5 - 8 = -7 \quad \text{X}$$

$$\rightarrow x_{22} : 8 - 5 - 5 = -2 \quad \text{X}$$

$\Rightarrow$ Solution is NOT optimal since
 some reduced cost < 0 .

$\Rightarrow$ Entering Variable : $x_{21} \quad (-7)$

$$\rightarrow \min(x_{12}, x_{23}) = 10 \quad (x_{12} = 10)$$

Init Feasible

Source	Dest.			Supply
	1	2	3	
1	10	20	20	50
2	0	0	20	20
Demand	10	20	40	

a.) $x_{21} + 10 \Rightarrow 10$

b.) $x_{11} - 10 \Rightarrow 0$

c.) $x_{13} + 10 \Rightarrow 30$

d.) $x_{23} - 10 \Rightarrow 10$

Source	Dest.			Supply
	1	2	3	
1	0	20	30	50
2	10	0	10	20
Demand	10	20	40	

→ new basic vars: $x_{21} = 10$ $x_{13} = 30$
 $x_{12} = 20$ $x_{23} = 10$

New Dual Vars:

Let $u_1 = 0$, then ...

$$\rightarrow x_{12} : 5 = u_1 + v_2 \Rightarrow v_2 = 5$$

$$\rightarrow x_{13} : 4 = u_1 + v_3 \Rightarrow v_3 = 4$$

$$\rightarrow x_{21} : 6 = u_2 + v_1$$

$$\rightarrow x_{23} : 9 = u_2 + v_2 = u_2 + 4 \Rightarrow u_2 = 5$$

$$\text{From } x_{21} : 6 = 5 + v_1 \Rightarrow v_1 = 1$$

New Dual Vars: $u_1 = 0, u_2 = 5$

$$v_1 = 1, v_2 = 5, v_3 = 4$$

• Check Optim:

$$x_{11} : 8 - 0 - 1 = 7$$

$$x_{22} : 8 - 5 - 5 = -2 \quad \text{X}$$

$\rightarrow$ Still not optimal, entering x_{22}

$\Rightarrow$ Entering Var. x_{22}

$$\rightarrow \min(x_{12}, x_{23}) = \min(20, 10) = 10$$

Source	Destination			Supply
	1	2	3	
1	0	10	40	50
2	10	10	0	20
Demand	10	20	40	

$\rightarrow$ new basic vars: $x_{12} = 10$, $x_{21} = 10$
 $x_{22} = 10$, $x_{13} = 40$

New Dual Vars.

Let $u_1 = 0$, then...

$$\rightarrow x_{12} : 5 = u_1 + u_2 \Rightarrow u_2 = 5$$

$$\rightarrow x_{13} : 4 = u_1 + u_3 \Rightarrow u_3 = 4$$

$$\rightarrow X_{21} : 6 = u_2 + v_1$$

$$\rightarrow X_{22} : 8 = u_2 + v_2 = u_2 + 5 \Rightarrow u_2 = 3$$

From $X_{21} : v_1 = 3$

New Dual Vars: $u_1 = 0, u_2 = 3$
 $v_1 = 3, v_2 = 5, v_3 = 4$

• Check Optimal:

$$\rightarrow X_{11} : 8 - 0 - 3 = 5$$

$$\rightarrow X_{23} : 9 - 3 - 4 = 2$$

$\Rightarrow$ All reduced costs $\geq 0 \rightarrow$ Optimal Found

Final Optimal Plan

Source	Destination		
	1	2	3
1	0	10	40
2	10	10	0

Final Cost:

$$Z = (0)(8) + (10)(5) + (40)(4) + (10)(6) + (10)(8)$$

$$\rightarrow Z = 350$$

2. Consider the following cost matrix for a transportation problem in which the objective is to minimize the total transportation cost.

Source	Destination				Supply
	1	2	3	4	
1	6	9	4	7	20
2	3	2	5	6	40
3	4	8	2	3	15
Demand	10	25	15	25	

- (a) (10 points) Set up the transportation tableau and find an initial basic feasible solution.

$$\rightarrow \text{Total Supply: } 20 + 40 + 15 = 75$$

$$\text{Total Demand: } 10 + 25 + 25 + 15 = 75$$

$\Rightarrow$ Balanced problem

NW Corner Rule

$$a.) (S_1, D_1) \quad \text{Supply} = 20 \quad \text{Demand} = 10$$

$$\rightarrow \text{alloc } \min(20, 10) = 10$$

$$\Rightarrow S_1 = 10 \quad D_1 = 0$$

$$b.) (S_1, D_2) \quad \text{Supply} = 10 \quad \text{Demand} = 25 \rightarrow \min = 10$$

$$\Rightarrow S_1 = 0, \quad D_2 = 15$$

c.) (S2, D2) Supply = 40 Demand = 15 $\rightarrow$ min = 15

$\Rightarrow S_2 = 25 \quad D_2 = 0$

d.) (S2, D3) Supply = 25 Demand = 15 $\rightarrow$ min = 15

$\Rightarrow S_2 = 10 \quad D_3 = 0$

e.) (S2, D4) Supply = 10 Demand = 25 $\rightarrow$ min = 10

$\Rightarrow S_2 = 0 \quad D_4 = 15$

f.) (S3, D4) Supply = 15 Demand = 15 $\rightarrow$ min = 15

Updated Tableau

Source	Destination				Supply
	1	2	3	4	
1	10	10	0	0	20
2	0	15	15	10	40
3	0	0	0	15	15
Demand	10	25	15	25	

$\Rightarrow$ Basic Vars: $6 = 3 + 4 - 1 \rightarrow$ Valid basic feasible solution

(b) (10 points) Beginning with the initial basic feasible solution in part (a), solve the problem using the transportation algorithm.

→ Compute Dual Values (u_i, v_j)

Solving for basic $C_{ij} = u_i + v_j$

Let $u_1 = 0$

$$\rightarrow x_{11} = 10 \Rightarrow 6 = u_1 + v_1 = 0 + v_1 \Rightarrow v_1 = 6$$

$$\rightarrow x_{12} = 10 \Rightarrow 9 = u_1 + v_2 = 0 + v_2 \Rightarrow v_2 = 9$$

$$\rightarrow x_{22} = 15 \Rightarrow 2 = u_2 + v_2 = u_2 + 9 \Rightarrow u_2 = -7$$

$$\rightarrow x_{23} = 15 \Rightarrow 5 = u_2 + v_3 = -7 + v_3 \Rightarrow v_3 = 12$$

$$\rightarrow x_{24} = 10 \Rightarrow 6 = u_2 + v_4 = -7 + v_4 \Rightarrow v_4 = 13$$

$$\rightarrow x_{34} = 15 \Rightarrow 3 = u_3 + v_4 = u_3 + 13 \Rightarrow u_3 = -10$$

Dual Values: $u_1 = 0, u_2 = -7, u_3 = -10$
 $v_1 = 6, v_2 = 9, v_3 = 12, v_4 = 13$

• Check Optimality:

$$\rightarrow x_{13} : 4 - 0 - 12 = -8 \quad \times$$

$$\rightarrow x_{14} : 7 - 13 = -6 \quad \times$$

$\Rightarrow$ Presence of reduced cost $< 0 \Rightarrow$ not optimal

Entering Var = $x_{13} = -8$ (most negative)

$$\rightarrow \min(x_{23}, x_{12}) = \min(15, 10) = 10$$

Source	Destination				Supply
	1	2	3	4	
1	10	0	10	0	20
2	0	25	5	10	40
3	0	0	0	15	15
Demand	10	25	15	25	

$$\rightarrow \text{Num. of basic vars} = 3 + 4 - 1 = 6$$

° Compute Dual Vars.

Let $u_1 = 0$, then ...

$$\rightarrow x_{11} = 10 \Rightarrow c_{11} = 6 = u_1 + v_1 \Rightarrow v_1 = 6$$

$$\rightarrow x_{13} = 10 \Rightarrow c_{13} = 4 = u_1 + v_3 \Rightarrow v_3 = 4$$

$$\rightarrow x_{22} = 25 \Rightarrow c_{22} = 2 = u_2 + v_2 \quad \text{Solving for } v_2$$

$$\rightarrow x_{23} = 5 \Rightarrow c_{23} = 5 = u_2 + v_3 = u_2 + 4 \Rightarrow u_2 = 1$$

$$\rightarrow x_{22} = 2 = 1 + v_2 \Rightarrow v_2 = 1$$

$$\rightarrow x_{24} = 10 \Rightarrow c_{24} = 6 = u_2 + v_4 = 1 + v_4 \Rightarrow v_4 = 5$$

$$\rightarrow x_{34} = 15 \Rightarrow c_{34} = 3 = u_3 + v_4 = u_3 + 5 \Rightarrow u_3 = -2$$

New Dual Vars: $u_1 = 0$, $u_2 = 1$, $u_3 = -2$
 $v_1 = 6$, $v_2 = 1$, $v_3 = 4$, $v_4 = 5$

Check Optimality:

$$° x_{12} = 9 - 0 - 1 = 8 \quad \checkmark$$

$$° x_{14} = 7 - 5 = 2 \quad \checkmark$$

$$° x_{21} = 3 - 1 - 6 = -4 \quad \times$$

→ Presence of reduced cost $x_{21} = -4 \Rightarrow$ not optimal

Entering Var $x_{21} = -4$

$$\rightarrow \min(x_{22}, x_{13}) = \min(25, 10) = 10$$

Source	Destination			
	1	2	3	4
1	20	0	0	0
2	10	15	15	10
3	0	0	0	15

• Compute Dual Vars

Let $u_1 = 0$, then...

$$\rightarrow x_{11} = 20 \Rightarrow 6 = u_1 + v_1 \Rightarrow v_1 = 6$$

$$\rightarrow x_{21} = 10 \Rightarrow 3 = u_2 + v_1 \Rightarrow u_2 = -3$$

$$\rightarrow x_{22} = 15 \Rightarrow 2 = u_2 + v_2 = -3 + v_2 \Rightarrow v_2 = 5$$

$$\rightarrow x_{23} = 15 \Rightarrow 5 = u_2 + v_3 = -3 + v_3 \Rightarrow v_3 = 8$$

$$\rightarrow x_{24} = 10 \Rightarrow 6 = -3 + v_4 \Rightarrow v_4 = 9$$

$$\rightarrow x_{34} = 15 \Rightarrow 3 = u_3 + 9 \Rightarrow u_3 = -6$$

New Dual Values: $u_1 = 0$, $u_2 = -3$, $u_3 = -6$

$$v_1 = 6, v_2 = 5, v_3 = 8, v_4 = 9$$

Check Optimality:

$$\rightarrow x_{12}: 9 - 5 = 4 \quad \checkmark$$

$$\rightarrow x_{13}: 4 - 8 = -4 \quad \times$$

$$\rightarrow x_{14}: 7 - 9 = -2 \quad \times$$

$\Rightarrow$ Still not optimal.

Entering Var $X_{13} = -4$

Source	Destination				Supply
	1	2	3	4	
1	20	0	0	0	20
2	10	15	15	10	50
3	0	0	0	15	15
Demand	10	25	15	25	

New Dual Vars: $u_1 = 0$, $u_2 = -3$, $u_3 = -6$
 $v_1 = 6$, $v_2 = 5$, $v_3 = 8$, $v_4 = 9$

Compute Dual Vars:

Let $u_1 = 0$, then...

$$\rightarrow x_{11} : 6 = u_1 + v_1 = 0 + v_1 \Rightarrow v_1 = 6$$

$$\rightarrow x_{13} : 4 = u_1 + v_3 = 0 + v_3 \Rightarrow v_3 = 4$$

$$\rightarrow x_{22} : 2 = u_2 + v_2 \Rightarrow u_2 \text{ from } \rightarrow x_{23} : 5 = u_2 + v_3 = u_2 + 4 \Rightarrow u_2 = 1$$

$$\rightarrow x_{22} : 2 = 1 + v_2 \Rightarrow v_2 = 1$$

$$\rightarrow x_{24} : 6 = u_2 + v_4 = 1 + v_4 \Rightarrow v_4 = 5$$

$$\rightarrow x_{34} : 3 = u_3 + v_4 = u_3 + 5 \Rightarrow u_3 = -2$$

New Dual Vars: $u_1 = 0, u_2 = 1, u_3 = -2$

$$v_1 = 6, v_2 = 1, v_3 = 4, v_4 = 5$$

Checking Optimality:

$$\rightarrow x_{12} : 9 - 0 - 1 = 8 \quad \checkmark$$

$$\rightarrow x_{14} : 7 - 0 - 5 = 2 \quad \checkmark$$

$$\rightarrow x_{21} : 3 - 1 - 6 = -4 \quad \times$$

$$\rightarrow x_{31} : 4 + 2 - 6 = 0 \quad \checkmark$$

$$\rightarrow x_{32} : 8 + 2 - 1 = 9 \quad \checkmark$$

$$\rightarrow x_{33} : 2 + 2 - 4 = 0 \quad \checkmark$$